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Project Report

On

Equations Knowledge Graph for Quantum Physics Papers

Submitted by

Satya Naga Vamsi Ganesh Manepalli

Course: **Natural Language Processing**

Supervised by

Prof. Dr. Patrick Levi

Ostbayerische Technische Hochschule Amberg-Weiden

Department of Electrical Engineering, Media and Computer Science

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1 Introduction

This project builds a prototype pipeline to extract a structured equation knowledge graph from arXiv quantum physics papers. For each enumerated equation the system captures: the LaTeX source, a natural-language meaning, definitions for all symbols, and graded pairwise relations to every other equation in the same paper. The final dataset contains 352 equations from 66 papers. All extraction is done from arXiv HTML only. No language model is used with prompting; embedding similarity is used solely for vector operations between phrase representations, which is explicitly permitted under the project rules.

2 Key Ideas

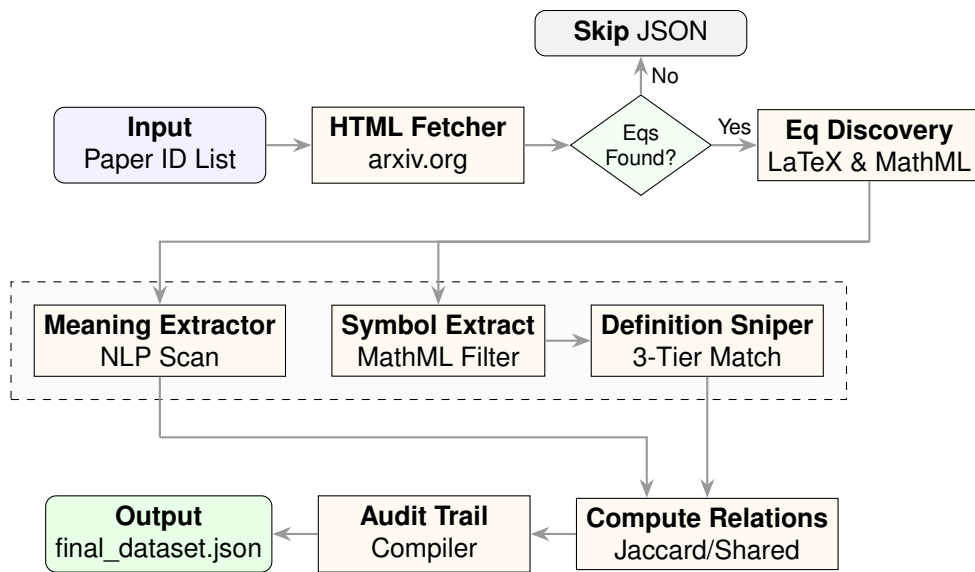


Figure 1: System architecture and extraction pipeline for generating the Equation Knowledge Graph.

2.1 Source: arxiv HTML over PDF or LaTeX

Papers are fetched from `arxiv.org` [1], which renders each arXiv paper as clean HTML with embedded MathML. The verbatim author LaTeX is stored in the `alttext` attribute of every `<math>` element, which is more reliable than re-extracting LaTeX from PDF renderings. Symbol-level access comes from MathML `<mi>` tokens, which already handle Unicode-to-LaTeX folding and function-name disambiguation (e.g. the token sequence `s-i-n` is recognised and dropped as a function name, not a symbol). A disk cache and a 3-second crawl delay respect arXiv's robots.txt.

2.2 Equation and Symbol Extraction

Enumerated equations are located via `<span class="ltx_tag_equation">` tags in the arxiv HTML. LaTeX is stitched from `<annotation encoding="application/x-tex">` child elements, handling multi-line aligned environments by concatenation. Symbol keys are extracted from MathML `<mi>` tokens after dropping function-name runs. Subscripted aliases (e.g. H_{eff}) are tracked separately for wider context matching.

2.3 Symbol Definition: Tiered Sniper with Memory Inheritance

Symbol definitions are extracted using a three-tier sniper that operates on a context window of up to ten sentences before and after each equation (Table 1).

Table 1: Definition extraction tiers, applied in priority order.

Tier	Strategy
1 – Trigger	Sentence contains “where”, “let”, “denotes”, or similar; regex captures the noun phrase following a copula. A spaCy [4] dependency parse is used as a fallback within this tier.
2 – Appositive / domain	“the Hamiltonian H ” (noun phrase directly before the symbol) or a domain constraint such as $X \in \{0, 1\}$.
3 – Bare adjacency	Nearest content word adjacent to the symbol; only fires when math context indicators are present.

Two cross-equation mechanisms extend coverage. **Memory inheritance**: a definition resolved for equation n is reused for equations $n+1$ and $n+2$ without re-searching, capturing derivation chains where the same symbols recur. **Pair propagation**: when two symbols always co-occur (e.g. creation/annihilation operators c , $c^\dagger$), a resolved definition propagates to the unresolved partner.

When no tier matches, an ensemble fallback counts adjacent and co-occurring noun phrases across all context sentences, applies glue-word penalties (“function”, “operator”, “value” are systematically over-represented and receive score multipliers < 1), and selects the most frequent content phrase. A dedicated summation-index rule independently tags letters appearing in $\sum \dots$ subscripts as *integer index / counter* before any text search is attempted, preventing noisy matches for $i, j, k, \dots$.

2.4 Equation Meaning Extraction

Meaning extraction applies four strategies in decreasing order of confidence. (1) A named-equation pattern matches phrases such as “Schrödinger equation” or “Bell inequality” anywhere in surrounding text. (2) A colon-introducer rule captures the last sentence before the equation if it ends with a colon, which authors use to introduce the equation being presented. (3) A governing-verb pattern captures clauses containing “is given by”, “reads”, “describes”, “is defined as”, and related phrases. (4) If spaCy is available, the three preceding sentences are scanned for verbs with defining lemmas. An empty string is returned rather than guessing when no evidence is found; this is preferable to a plausible but wrong meaning in a knowledge graph.

2.5 Equation Relations

Every pair of equations within a paper is classified into one of three grades. *Strong*: at least two shared symbol keys, or the normalised LaTeX strings are structurally equivalent by 3-gram Jaccard similarity (≥ 0.85). *Potential*: exactly one shared symbol, or Jaccard ≥ 0.4 . *None*: otherwise. Strong relations include a short description (e.g. “shares variables: H , ψ ”, “special case of general form”). This conservative structural approach avoids false positives while reliably identifying the most obvious mathematical dependencies between equations.

3 Generalization

Three design decisions support generalization beyond the assigned paper list. First, no paper-specific thresholds are hard-coded. The trigger vocabulary (“where”, “let”, “denotes”, etc.) follows LaTeX prose conventions universal across arXiv preprints. Second, both the sniper and the meaning extractor fail gracefully: empty strings are returned when evidence is absent, rather than fabricating output. Third, the approach handles all seven quantum paper sub-types identified in corpus analysis (circuit optimization, variational, error correction, ZX-calculus, complexity, simulation, and cryptography) because it relies on HTML structure and prose patterns rather than domain-specific heuristics.

The known hard cases are ZX-calculus papers, where equations are circuit diagrams embedded as figures rather than algebra, and optical/measurement papers where symbol definitions appear only in figure captions. Neither caption text nor figures are currently scanned. Resolving these would require figure-caption parsing or image understanding, both exceeding the no-prompting constraint.

4 Quality

Table 2 summarises extraction quality across the 352-equation prototype dataset (66 papers, of which 54 contained enumerated equations). The strongest results occur in complexity and error-correction papers, which follow strict definition-before-use conventions matching the Tier 1 trigger patterns closely. The weakest occur in papers that define symbols in non-standard phrasing (e.g. “taking H to be. . .”) or that lack a dedicated Preliminaries section.

Table 2: Extraction quality on the 352-equation prototype dataset (66 papers).

Metric	Result	Notes
Total equations extracted	352	target: 350–356
Symbol definitions filled	57%	1,118 / 1,947 symbols
of which GOOD (≥ 2 words)	29%	575 symbols
of which WEAK (1 word)	27%	543 symbols
of which EMPTY	42%	829 symbols
Equation meanings filled	84%	298 / 352 equations
of which ≥ 15 words (informative)	57%	171 meanings
Relation pairs graded strong	44%	912 / 2,030 pairs
Relation pairs graded potential	28%	574 / 2,030 pairs
Papers with no enumerated equations	18%	12 / 66 papers
Audit trail entries per equation (avg)	9.5	min 4, max 25

Three residual limitations cannot be eliminated within the rule-based constraint. **Ambiguous and high-frequency single-letter symbols:** letters such as e , W , P have low definition rates (28%, 23%, 25% respectively) because they carry different meanings across papers and across equations within the same paper. The memory inheritance mechanism partially addresses this but cannot resolve true ambiguity. Notably, standard symbols `phi` and `hbar` show 0% definition rate in this dataset — these should be covered by a pre-defined dictionary and represent a concrete improvement point for future work. **Implicit definitions:** symbols defined only in referenced papers are left empty (flaggable but not resolvable). **Relation semantics:** the grading captures structural relatedness but not semantic intent, so a “potential” grade may cover both a true derivation step and a coincidental shared variable. The 84% meaning extraction

rate and 9.5 average audit trail entries per equation indicate that the evidence chain is well documented even when definitions are missing.

5 Conclusion

The pipeline demonstrates that a tiered linguistic sniper, combined with memory inheritance across equations and glue-word penalization, can extract a useful equation knowledge graph from arXiv HTML without any prompting. The audit trail in the JSON output records the exact evidence sentence and method name for every extraction step, enabling transparent quality assessment and demonstrating full compliance with the project constraints.

Declaration on AI Tool Use

AI assistants (ChatGPT, Claude) were used during development to analyze representative arXiv papers and to assist with iterative code refactoring. All extraction logic, decision rules, and evaluation results were authored and validated by the student. No AI model is invoked at runtime in the submitted system: no prompts are issued and no generated text appears in the output dataset.

References

- [1] arxiv.org (2024). HTML5 rendering of arXiv preprints. <https://arxiv.org>
- [2] Richardson, L. (2007). Beautiful Soup. <https://www.crummy.com/software/BeautifulSoup/>
- [3] W3C (2014). Mathematical Markup Language (MathML) Version 3.0, Second Edition. <https://www.w3.org/TR/MathML3/>
- [4] Honnibal, M. and Montani, I. (2017). spaCy: Industrial-strength natural language processing. <https://spacy.io>